PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Clinical efficacy and security of glycyrrhizic acid preparation in the treatment of anti-SARS-CoV-2 drug-induced liver injury: a protocol of systematic review and meta-analysis
AUTHORS	Tian, Xia; Gan, Wenfan; Nie, Yisen; Ying, Rongtao; Tan, Yongji; Chen, Junli; Chen, Mei; zhang, chuantao

VERSION 1 – REVIEW

REVIEWER	Yue, Shi-Jun Shaanxi University of Chinese Medicine
REVIEW RETURNED	30-Mar-2021

GENERAL COMMENTS	1. It is difficult to figure out whether the liver injury mentioned in the title refers to the patient's pre-existing disease (e.g., cirrhosis, hepatocellular carcinoma, non-alcoholic fatty liver disease, autoimmune liver diseases or liver transplant) or a complication after contracting SARS-CoV-2. 2. In the inclusion criteria, one of the requirements for the type of participant is "patients with liver injury (> 18 years old) with COVID-19", but it lacks a certain degree of strictness. Also, the age distribution of the susceptible population of patients with COVID-19 is different. 3. In the exclusion criteria, liver injury resulting from toxicity from commonly used drugs in COVID-19 patients should be included. 4. The "TCM symptom score" mentioned in the primary outcome lacks correlation with the Inclusion criteria. 5. Since liver cirrhosis reportedly increased length of stay, it is strongly to consider length of stay as one of the measured outcomes.

REVIEWER	Sahoo, Suban SV National Institute of Technology, Department of Chemistry
REVIEW RETURNED	03-Apr-2021

GENERAL COMMENTS	1. 'In view of the serious epidemiological situation of COVID-19, there is no effective treatment at present'. I think vaccines as well as FDA approved drug for COVID-19. So, this sentence need some modification. 2. Author should cite the computation work, where the GAP has been proposed as potential phytochemicals against COVID-19 (Computers in Biology and Medicine 124 (2020) 103936).

VERSION 1 – AUTHOR RESPONSE

Reviewer #1:
1. It is difficult to figure out whether the liver injury mentioned in the title refers to the patient's pre-existing disease (e.g., cirrhosis, hepatocellular carcinoma, non-alcoholic fatty liver disease, autoimmune liver diseases or liver transplant) or a complication after contracting SARS-CoV-2.

Response: We have revised the title to "Clinical efficacy and security of glycyrrhizic acid preparation in the treatment of anti-SARS-CoV-2 drug-induced liver injury: a protocol of systematic review and meta-analysis". So, we will study the anti-SARS-CoV-2 drug-induced liver injury in paitents with COVID-19.

2. In the inclusion criteria, one of the requirements for the type of participant is "patients with liver injury (> 18 years old) with COVID-19", but it lacks a certain degree of strictness. Also, the age distribution of the susceptible population of patients with COVID-19 is different.

Response: Because the population is generally susceptible to SARS-CoV-2, there is no restriction on the age of patients with COVID-19.

3. In the exclusion criteria, liver injury resulting from toxicity from commonly used drugs in COVID-19 patients should be included.

Response: In the exclusion criteria, we included liver injury resulting from toxicity from commonly used drugs in COVID-19 patients. The patients with pre-existing liver disease (e.g., cirrhosis, hepatocellular carcinoma, non-alcoholic fatty liver disease, etc.) will be excluded.

4. The "TCM symptom score" mentioned in the primary outcome lacks correlation with the Inclusion criteria.

Response: We deleted the "TCM symptom score" in the inclusion criteria.

5. Since liver cirrhosis reportedly increased length of stay, it is strongly to consider length of stay as one of the measured outcomes.

Response: We included the length of stay as the primary outcome.

Reviewer #2:
1. 'In view of the serious epidemiological situation of COVID-19, there is no effective treatment at present'. I think vaccines as well as FDA approved drug for COVID-19. So, this sentence needs some modification.

Response: We modified this sentence with "COVID-19 is in the global pandemic. The vaccine is the key to the prevention and control of this disease. Current clinical studies confirmed that the vaccine is safe and effective with few adverse reactions, but the vaccine can only reduce the risk of SARS-CoV-2 infection. There is insufficient clinical evidence to confirm that anti-SARS-CoV-2 drugs(e.g. alpha-interferon, lopinavir/ritonavir, ribavirin, and chloroquine phosphate) are effective in suppressing the virus, which also cause a variety of adverse effects such as diarrhea, elevated transaminases, and skin rash. One study consolidated and reviewed the available clinical and preclinical relevant results, showing mixed results in terms of efficacy within the framework of current clinical protocols.

Therefore, it is necessary to find an effective and safe scheme for the treatment of anti-SARS-CoV-2 drug-induced liver injury."

2. Author should cite the computation work, where the GAP has been proposed as potential phytochemicals against COVID-19 (Computers in Biology and Medicine 124 (2020) 103936).

Response: We cited this article in introduction section.

Once again, thank you very much for your constructive comments and suggestions which would help us both in English and in depth to improve the quality of the paper.

VERSION 2 – REVIEW

REVIEWER	Yue, Shi-Jun Shaanxi University of Chinese Medicine
REVIEW RETURNED	08-Jun-2021

GENERAL COMMENTS	The authors have addressed all the questions that I have raised and the new submission is now acceptable.